

SIGNIVERSE: AN EDUCATIONAL GAME FOR LEARNING JORDANIAN SIGN LANGUAGE (LIU) THROUGH GESTURE RECOGNITION



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ABSTRACT

Deaf and Hard-of-Hearing (HoH) children in Jordan lack self-based, interactive resources for learning Jordanian Sign Language (LIU). This project introduces SigniVerse, an educational serious game that integrates sign recognition models to support Jordanian Sign Language (LIU) learning through interactive gameplay and real-time feedback. The game focuses on teaching foundational categories, including letters, numbers, and everyday signs through minigames. The use of computer vision enables immediate feedback, allowing users to self-correct and learn at their own pace.

Exploratory interviews were conducted with one DHH individual and one DHH trainer. The interviews highlighted the importance of early-age learning, and informed selection of the signs.

The system is designed for scalability, able to accommodate the addition of new vocabulary and minigames. Furthermore, the custom dataset will be made publicly available to support future research.

This study evaluates the usability, engagement, and technical performance of the game rather than its educational outcomes, which are reserved for future work.

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A special thanks goes to trainers and members of the Deaf and Hard of Hearing community who were eager to share their resources and their experiences to help us make the application impactful and meaningful. Your contribution is truly appreciated.

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LIST OF ABBREVIATIONS

Deaf and Hard-of-Hearing	DHH
Data Flow Diagram	Data Flow Diagram
Jordanian Sign Language	LIU
Teachers of the Deaf and Hard of Hearing	TDHH

1.0 INTRODUCTION

1.1 Overview

This project focuses on developing an interactive educational game designed to support deaf and hard-of-hearing (DHH) children in learning sign language in an engaging and accessible way. The game integrates computer vision technology to recognize hand gestures and provide real-time feedback, allowing children to practice sign-language movements accurately and independently.

Through a combination of visual tutorials, animations, and interactive challenges, the game aims to create a fun and motivating learning environment that aligns with the visual learning style of DHH children. By leveraging game-based learning and modern vision technology, the project offers an innovative solution to support inclusive education for young learners.

1.2 Problem Statement

Traditional Jordanian Sign Language (LIU) learning environments are often restricted to specialized centers, therefore making it difficult for children to practice independently outside of a classroom. Existing digital solutions are largely static (video-based), lacking the interactivity required to ensure the child practices the sign correctly.

1.3 Project Aim and Objectives

1.3.1 Project aim:

The aim of this project is to develop an interactive educational game designed to support deaf and hard-of-hearing (DHH) children who are learning Jordanian sign language in an engaging, visual, and accessible way. The game seeks to provide an enjoyable learning experience that supports independent practice and reinforces language development.

1.3.2 Project Objectives:

- To design and develop an interactive serious game that supports Jordanian Sign Language (LIU) practice for children aged 7-9 through engaging minigames and a reward system.
- To collect and publish a labelled dataset of LIU signs for model training and supporting LIU research.
- To design a modular, scalable game architecture that supports the addition of new vocabulary and categories.
- To evaluate the game through user-based testing focusing on UI quality, usability, engagement, responsiveness, and user satisfaction.

1.4 Main Motivations

The main motivation behind this project is to provide a useful resource for Deaf and Hard of Hearing children in Jordan.

1.5 Project Scope

The system will be developed as an interactive educational game designed to teach sign language to children aged between 7 and 9 years, including both deaf and hearing children. The game will utilize artificial intelligence and computer vision technologies to recognize hand gestures and provide real-time feedback, creating an engaging and inclusive learning experience.

The application aims to enhance early sign language learning through fun, interactive, and visual activities that support communication skills and cognitive development for young children.

The game will provide the following key features:

- Interactive learning games focused on teaching basic sign language letters, numbers, and words.
- Real-time hand gesture recognition using computer vision and artificial intelligence.
- Instant feedback to guide children and reinforce correct sign performance.
- Age-appropriate game design.
- Support for both deaf and non-deaf children to promote inclusive learning.
- Simple and child-friendly user interface with visual instructions.
- Educational content presented in a fun and engaging manner to maintain children's attention.

1.6 Project Limitations

One of the major limitations faced in this project is the lack of suitable and high-quality sign-language datasets for training the computer vision model. Most of the available datasets are either in English sign language or in various Arabic dialects that do not match

the Jordanian Sign Language required for this project. This mismatch makes it challenging to train a model that can accurately recognize and evaluate the specific gestures used by Jordanian DHH children.

Furthermore, creating a custom dataset that reflects the correct Jordanian signs requires extensive time, resources, and support from sign-language experts. As a result, the gesture-recognition accuracy may be limited, which can affect the reliability of the real-time feedback provided to the player during the learning process.

1.7 Project Expected Output

The expected output of this project is an interactive educational game that enables deaf and hard-of-hearing (DHH) children to learn sign language using engaging visual activities supported by computer vision technology. The game will provide sign-language lessons, practice exercises, and real-time feedback, allowing users to practice signs while the system analyzes their hand movements and gestures through computer vision.

The final product is expected to include:

- A fully functional educational game with a child-friendly interface.
- Integrated computer vision features capable of detecting and evaluating hand gestures.
- Visual tutorials demonstrating correct sign-language gestures.

1.8 Report Outline

- Chapter One: Introduction to the project, including motivation, problem statement, aims, objectives, scope, limitations, and schedule.
- Chapter Two: Review of related existing systems, the overall problem of existing systems, and the solution.
- Chapter Three: Feasibility study, requirements engineering and specification, and the usability and user experience goals.
- Chapter Four: System architecture and design, represented by diagrams and a graphical user interface.
- Chapter Five: System implementation, including the high-fidelity prototype, and any details pertaining the implementation tools used.
- Chapter Six: System deployment and evaluation, including system and user testing, and a glimpse of the tutorial.
- Chapter Seven: Conclusion and future work, summarizing project achievements and future directions.

1.9 Project Schedule

Table 1: Project Schedule

Task ID	Task Description	Duration (Days)	Dependency	Responsible Partner(s)	Resources
T1	Deep research about deaf people, educational game and deep leaning	7	none	All team	Papers, online videos, and Internet research
T2	Data collection (alphabet signs)	7	T1	All team	Papers and Recorded videos
T3	Developing the model	7	T2	Farah	Recorded videos and professor consultations
T4	Testing the model	7	T3	Farah	Professor consultations
T5	Defining the scope and targeted people	5	T1	Farah, Yara	Papers, internet research and meeting deaf people and translators
T6	Writing CH1: Introduction	5	T5	Farah, Eman	Papers, internet research and meeting DHH people and translators
T7	Writing CH2: Related Existing Systems	5	T5	Yara, Rama	Papers, Internet research and meeting DHH people and translators
T8	Continue dataset collection (number signs)	7	T4	All team	Papers and Recorded videos
T9	Train the numbers model	7	T8	Farah	Recorded videos and professor consultations
T10	Defining the categories in the game	7	T5	Yara	Sign language references
T11	Defining the game loop	3	T10	All team	game development process
T12	Writing CH3	7	T11	All team	Papers, Internet research and

					meeting DHH people and translators
T13	Writing CH4	7	T11	All team	Papers, Internet research and meeting DHH people and translators
T14	Designing the prototype	5	T11	Rama	Figma
T15	Develop game to test FastAPI integration	7	T14	Farah, Yara	Internet research
T16	Train the alphabet model	5	T2	Farah	Recorded videos and professor consultations
T17	Design game assets	3	T16	Rama	Figma
T18	Develop number signs game	14	T17	Eman	Internet research
T19	Develop alphabet signs game	14	T17	Rama	Internet research
T20	Hosting on Render	3	T18, T19	Yara	Internet research
T21	Improving game UI & mechanics	3	T20	All team	
T22	Continue dataset collection (household signs)	5	T21	All team	Papers and Recorded videos
T23	Train the household category model	7	T22	Farah	Recorded videos and professor consultations
T24	Design household signs game assets	5	T22	Rama	Figma
T25	Develop household signs game	14	T24	Yara	Internet research
T26	Testing	7	T25	All team	Meeting children to test game
T27	Refining game	7	T26	All team	Feedback from testing
T28	Finalizing documentation	5	T27	All team	



Figure 1: Gantt Chart

1.10 Project Software and Hardware Requirements

1.10.1 Software Requirements

Table 2: Software Requirements

Component	Software	Description
Operating System	Windows 11	Development and testing environment.
Programming Language	Python	AI and computer vision processing.
Game Engine	Unity	Game development.
Computer Vision	OpenCV	Image processing.
Hand Tracking	MediaPipe	Hand landmark detection.
AI Model	LSTM	Gesture recognition.
AI Framework	TensorFlow	Model training and deployment.
IDE	VS Code	Coding and debugging.
Version Control	GitHub, google docs	Source code management.
UI/UX Design	Figma	Interface design.
IDE	Google colab and VS code	Used for writing, debugging, and testing the source code.

1.10.2 Hardware Requirements

Table 3: Hardware Requirements

Component	Specification	Description
Computer / Laptop	Intel i5 and equivalent	Used for development and testing.
RAM	16 GB (minimum)	Required for running AI tools smoothly.
Storage	500 GB SSD	Used for software and datasets.
Web Camera	720p or higher	Used for hand gesture capture.
GPU	NVIDIA GPU	Faster AI model training.
Speakers	Built-in or external	Audio feedback in the game.

2.0 Related Works

The field of gesture recognition for sign language has evolved significantly with the advancement of deep learning architectures. Rastgoo et al. (2021) provided a comprehensive survey of these applications, which highlighted the efficacy of Recurrent Neural Networks (RNNs) for temporal video analysis, alongside object detection frameworks such as Faster R-CNN, YOLO, and SSD for spatial feature extraction. This technical consensus supports the integration of Long Short-Term Memory (LSTM) networks - a specialized type of RNN - which has been successfully implemented in modern systems such as SIGNIFY (Ulrich et al., 2024) and PopSignAI (Starner et al., 2021).

While numerous educational applications exist for sign language, a significant portion lacks an integrated computer vision component, relying instead on static imagery or passive video playback. SIGNIFY introduced a gamified approach to teaching the manual alphabet; however, its scope remains limited to a small subset of signs. The most comparable architecture to our proposed system is PopSignAI, which utilizes real-time gesture recognition to facilitate learning.

SigniVerse distinguishes itself from existing solutions in two primary ways:

1. **Linguistic Localization:** SigniVerse is specifically designed for Jordanian Sign Language (LIU), addressing a critical local resource gap.
2. **Gameplay Diversity:** While existing prototypes often rely on a single, repetitive interaction loop, SigniVerse incorporates multiple minigames across distinct categories. This approach is designed to maintain high user engagement and prevent learning fatigue.

3.0 System Requirements Engineering and Analysis

3.1 Feasibility Study

3.1.1 Technical Feasibility

The project is feasible because Unity can capture the camera stream and send MediaPipe hand-landmark sequences to a Python API where the LSTM model classifies the sign. The API returns the predicted sign and confidence to Unity in real time, making integration simple and allowing easy model updates without rebuilding the game.

3.1.2 Operational Feasibility

The project is operationally feasible because it fits real learning settings (home/school) and delivers learning through gameplay. The child learns each sign through short interactive lessons, then practices and is assessed through mini-games (challenges, quizzes, and level-based tasks). The system provides instant feedback (correct/try again) with rewards, which increases motivation and supports continuous learning. It requires only a camera device and simple guidance for lighting and hand positioning.

3.1.3 Schedule Feasibility

The project is feasible in 6 months since the AI model is ready and the remaining tasks are building the Unity educational game and integrating it via a Python API. The remaining time will be used for testing, improving accuracy, polishing gameplay, and final documentation.

3.1.4 Legal Feasibility

The project is legally feasible as long as children's privacy is protected. This includes obtaining guardian consent for any data collection, minimizing stored data (preferably saving only hand landmarks), avoiding storing faces or videos, securing any stored information, and ensuring the game content is child-safe with properly licensed assets.

3.1.5 Economic Feasibility

The project is economically feasible because it can be developed at low cost using mostly free or open-source tools and existing hardware. Unity and the AI stack (Python, MediaPipe, OpenCV) can be used without significant licensing fees for academic development. The main cost is the team's time and effort for data collection, labeling, training, and testing. During development, the system can run locally, which keeps expenses minimal. If the project is published, additional costs may apply, such as developer account fees for releasing the app on Google Play (one-time registration) and the Apple App Store (annual developer subscription). Also, if the AI component is deployed through a hosted Python API, there may be hosting and maintenance costs. Overall, the expected expenses remain relatively low compared to the educational value and the potential for future adoption by schools and learning centers.

3.2 Requirement Gathering Techniques

Requirement Gathering is a critical part of the requirements management and definition process, with the goal of ensuring all requirements accurately reflect stakeholder needs.

For optimal results, several requirement gathering techniques were employed:

1. Stakeholder Interviews:

- a. Interviews with Deaf and Hard of Hearing (DHH) individuals

Interviews with individuals from the DHH community were conducted to gain insight into their lived experiences and educational hurdles. These discussions

focused on identifying gaps in current educational tools and learning difficulties faced by young DHH individuals.

b. Interviews with Teachers of the Deaf and Hard of Hearing (TDHH)

Consultations with TDHH were instrumental in defining the project's scope, shifting the project to focus on high-impact vocabulary to maximize the educational benefits of the target demographic, and aiming to improve their quality of life.

2. Document Analysis

A comprehensive review of literature regarding DHH individuals was conducted. Analysis of the literature revealed that DHH individuals tend to have lower academic scores, and attributed this trend not to cognitive ability, but to the ineffectiveness of traditional teaching methods. This finding validates the need for an interactive learning tool, specialized for DHH individuals.

3. Expert Consultation

Consultations with computer vision experts and developers of Sign Language systems were held regularly to help guide the selection of data acquisition methods and model architecture.

3.3 Targeted Users

The system will be developed as an interactive educational game designed to teach sign language to children aged between 7 and 9 years, including both deaf and hearing children.

3.4 Functional Requirements

Table 4: Functional Requirements

FR ID	Functional Requirement	Description
Game Launch & Navigation		
FR-1	Main Start Screen	The system shall display a main start screen when the game is launched; the start screen shall contain a single "Start" button.
FR-2	Start Game Button	The system shall allow the player to start the game using a "Start" button.
FR-3	Display Visual Theme	The system shall present a space-themed educational environment consistent with the SigniVerse concept.
FR-4	Display Sign Language Elements	The system shall display sign language symbols and hand gestures as part of the educational game interface.
FR-5	Button Interaction Feedback	The system shall provide simple visual and audio feedback when a button is clicked.
Open World Navigation		
FR-6	Space-Themed Open World	The system shall provide an open-world space-themed environment.
FR-7	Horizontal World Navigation	The system shall allow the player to navigate horizontally through the game world to explore available planets.
FR-8	Forward and Backward Planet Browsing	The system shall allow the player to move right to discover new planets and move left to return to previously visited planets.
FR-9	Unrestricted World Exploration	The system shall allow the player to explore the world without restrictions; this ensures that all learning content is available from the beginning of the game.
FR-10	No Enforced Learning Path	The system shall not enforce a predefined learning path or levels.
FR-11	Location Entry Interaction	The system shall allow the player to enter a learning area by colliding with a location object.
FR-12	Interactive Sign Language Learning Gameplay	The system shall load an interactive sign language learning activity when the player enters any learning location in the game world.

FR-13	Return to Main World	The system shall allow the player to exit a learning area and return to the main world.
Sign Language Learning Interaction		
FR-14	Display Target Sign	The system shall display a target sign language gesture to the player.
FR-15	Sign Demonstration	The system shall provide a visual demonstration of the sign.
FR-16	Gesture-Based Sign Performance	The system shall allow the player to perform the sign using their hands.
FR-17	Camera-Based Hand Tracking	The system shall capture hand movements using the device camera.
FR-18	Gesture Comparison	The system shall compare the player's gesture with the target sign.
FR-19	Gesture Validation	The system shall determine whether the performed sign is correct or incorrect.
Feedback, Points & Motivation		
FR-20	Immediate Learning Feedback	The system shall provide immediate feedback after each performed sign.
FR-21	Positive Reinforcement	The system shall use positive visual and audio feedback for correct signs.
FR-22	Supportive Error Feedback	The system shall encourage the player when an incorrect sign is performed.
FR-23	Point Reward System	The system shall award points for each correctly performed sign.
FR-24	Points Display	The system shall display the total points earned by the player.
FR-25	Progress Tracking	The system shall track and record the player's learning progress throughout the game.
FR-26	Non-Restrictive Scoring	The system shall allow point accumulation without locking or unlocking content.
Help & Error Handling		
FR-27	Sign Retry Capability	The system shall allow the player to retry the same sign multiple times.
Exit & Progress Saving		
FR-28	Game Exit	The system shall allow the player to exit the game at any time.
FR-29	Progress Saving	The system shall save the player's progress and learned signs.

FR-30	Progress Restoration	The system shall restore saved progress when the game is restarted.
Shop/Avatar System		
FR-31	Access In-Game Shop	The system shall allow the player to access the in-game shop from the level selection screen.
FR-32	Avatar Selection	The system shall allow the player to browse and select available spacecraft avatars.
FR-33	Avatar Purchase	The system shall allow the player to purchase avatars using earned stars.
FR-34	Avatar Customization	The system shall allow the player to equip a selected avatar for gameplay.
FR-35	Star Currency Usage	The system shall use accumulated stars as the in-game currency for purchases.
Numbers Mini-Game		
FR-36	Educational Mini-Games	The system shall provide educational mini-games related to sign language learning.
FR-37	Performance Rating	The system shall evaluate the player's performance and award a star rating upon completion of a game level.
FR-38	Interactive Object Collection	The system shall allow the player to collect or interact with game objects during gameplay.
Alphabets Mini-Game		
FR-40	Spaceship Control	The system shall allow the player to control a spaceship during gameplay activities.
FR-41	Obstacle Avoidance Challenge	The system shall present obstacles that the player must avoid while progressing through the game.
FR-42	Collectible Rewards	The system shall provide collectible stars that increase the player's score.
FR-43	Score Tracking	The system shall continuously track and display the player's current score during gameplay.
FR-44	Dynamic Game Environment	The system shall display a dynamic space environment containing moving objects, collectibles, and obstacles.
Home Puzzle Mini-Game		
FR-40	Home World Puzzle Activity	The system shall provide an interactive puzzle activity based on a virtual house environment.
FR-41	Four-Room House Representation	The system shall display a house consisting of four rooms: Bedroom, Living Room, Bathroom, and Kitchen.
FR-42	Room Identification	The system shall allow the player to identify and interact with the Bedroom, Living Room, Bathroom, and Kitchen.
FR-43	Puzzle Piece Placement	The system shall allow the player to place puzzle elements in their correct room locations.
FR-44	Room Validation	The system shall verify whether the selected room or puzzle placement is correct.
FR-45	Puzzle Completion Feedback	The system shall provide feedback upon successful completion of the house puzzle.
FR-46	Home Vocabulary Learning	The system shall teach home-related Jordanian Sign Language vocabulary through the Bedroom, Living Room, Bathroom, and Kitchen environments.

3.5 Non-functional Requirements

Table 5: Non-functional Requirements

NFR	Requirement	Description	Priority
NFR1	Performance	The game should load pages and provide real time update (e.g. giving feedback on the performance of the sign if it is true or not within 3-5 seconds).	5
NFR2	Security	All user data including stages of progress must be securely encrypted and stored.	5
NFR3	Availability	The game must maintain an operating rate of 99.9% at all times.	5
NFR4	Usability	The game should have a user-friendly interface.	4
NFR5	Compatibility	The system should support devices like IOS and Android	4

3.6 Domain Requirements

Table 6: Domain Requirements

DR	Description
DR1	Continuous Updates: regularly update and expand the content.
DR2	System Scalability: The system should handle an increasing number of users and transactions smoothly.

3.7 Usability and User Experience Goals

- **Ease of use:** the game should be easy to use and understandable without prior instructions.
- **Child-friendly interaction:** the game should be suitable for children with simple navigation and interactive elements.
- **Engagement and motivation:** the game should use rewards and scores to motivate children to continue learning.
- **Consistent design:** the game should maintain a consistent layout, colors, and interaction style throughout all screens.

4.0 System Design

4.1 Context Diagram

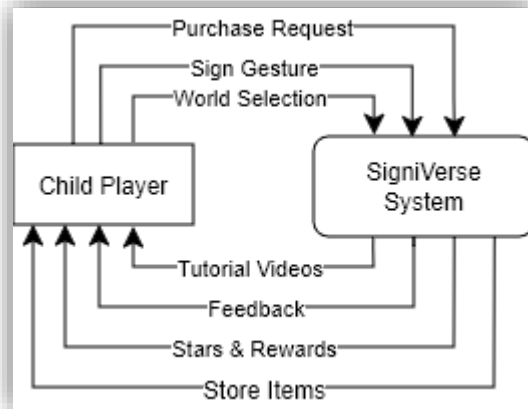


Figure 2: Context Diagram

4.2 Data Flow Diagram (DFD)

DFD Level 1:

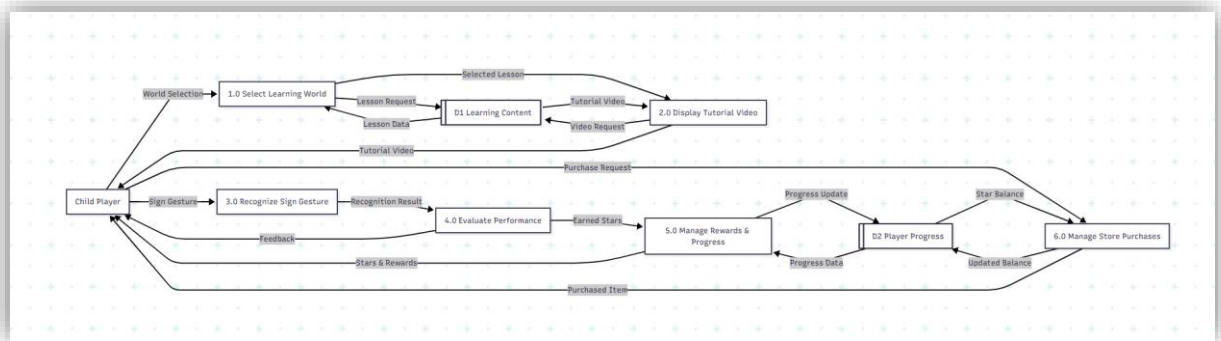


Figure 3: DFD Level 1

DFD Level 2:

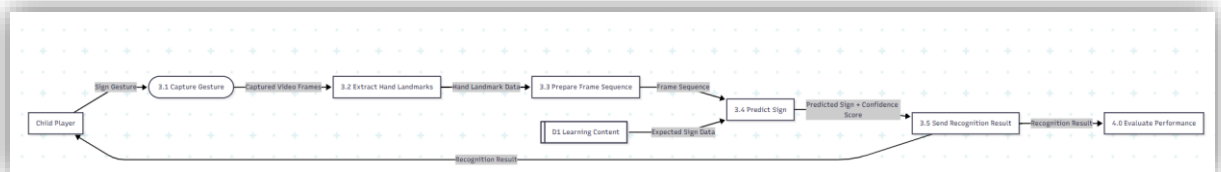


Figure 4: DFD Level 2

4.3 UML Use Case Diagram

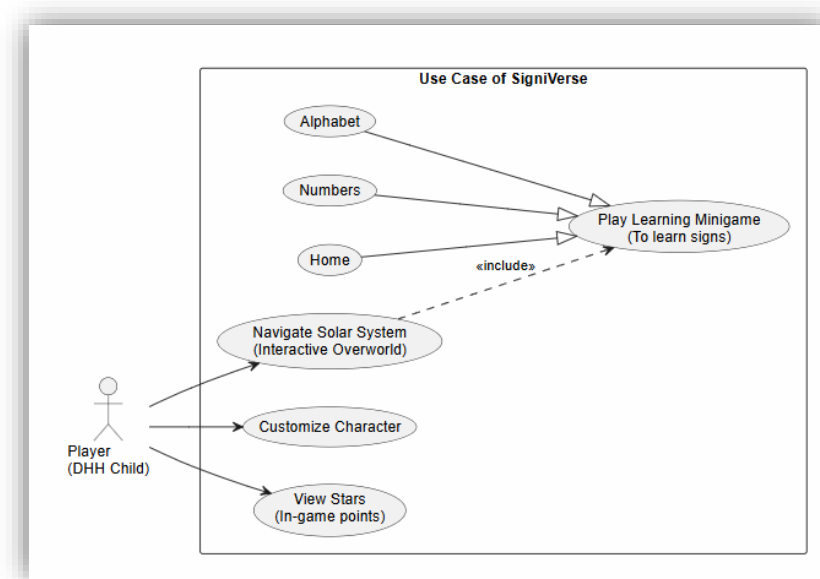


Figure 5: Player Use Case Diagram

4.4 UML Sequence Diagram

Table 7: Game Launch and Navigation Sequence Diagram details

Item	Details
Actors	Player
Description	Player launches the game and starts the SigniVerse experience from the main start screen.
Data	Start button, visual theme, sign language elements, button feedback
Stimulus	Player opens the game and clicks the Start button.
Response	The system displays the start screen, provides interaction feedback, and loads the game world.
Comments	Represents the first interaction between the player and the game.

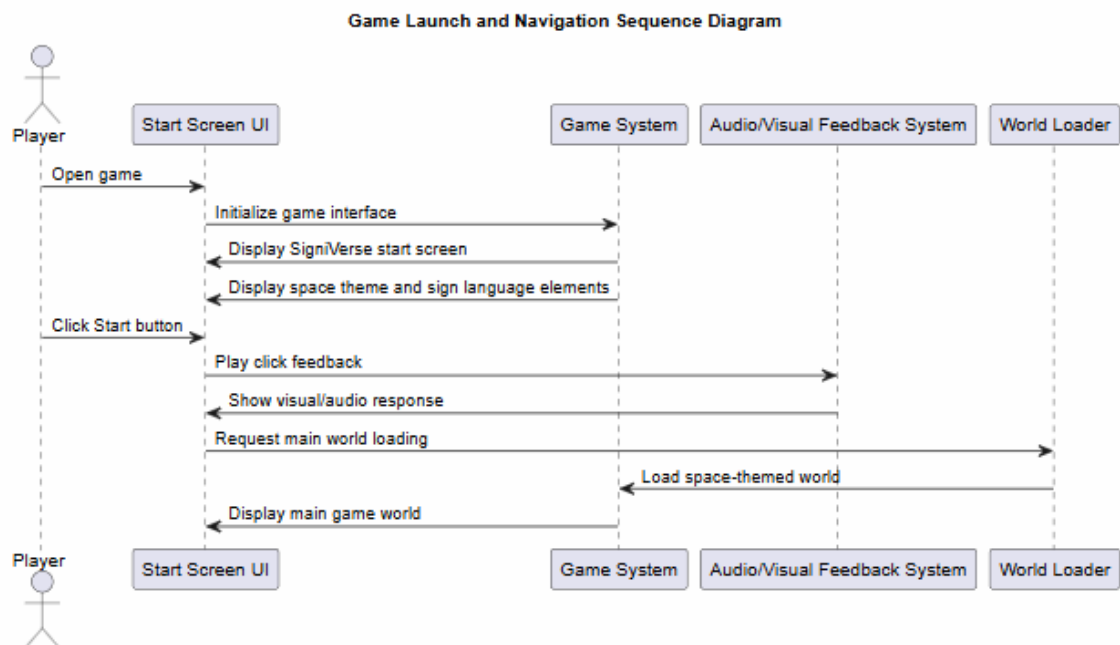


Figure 6: Game Launch and Navigation Sequence Diagram

Table 8: Open World Navigation Sequence Diagram details

Item	Details
Actors	Player
Description	Player navigates horizontally between planets to explore available learning worlds.
Data	Planet list, current planet position, movement direction, selected learning area
Stimulus	Player moves right or left, or collides with/selects a learning location.
Response	The system displays next or previous planets and loads the selected learning activity.
Comments	Supports free exploration without enforcing a fixed learning path.

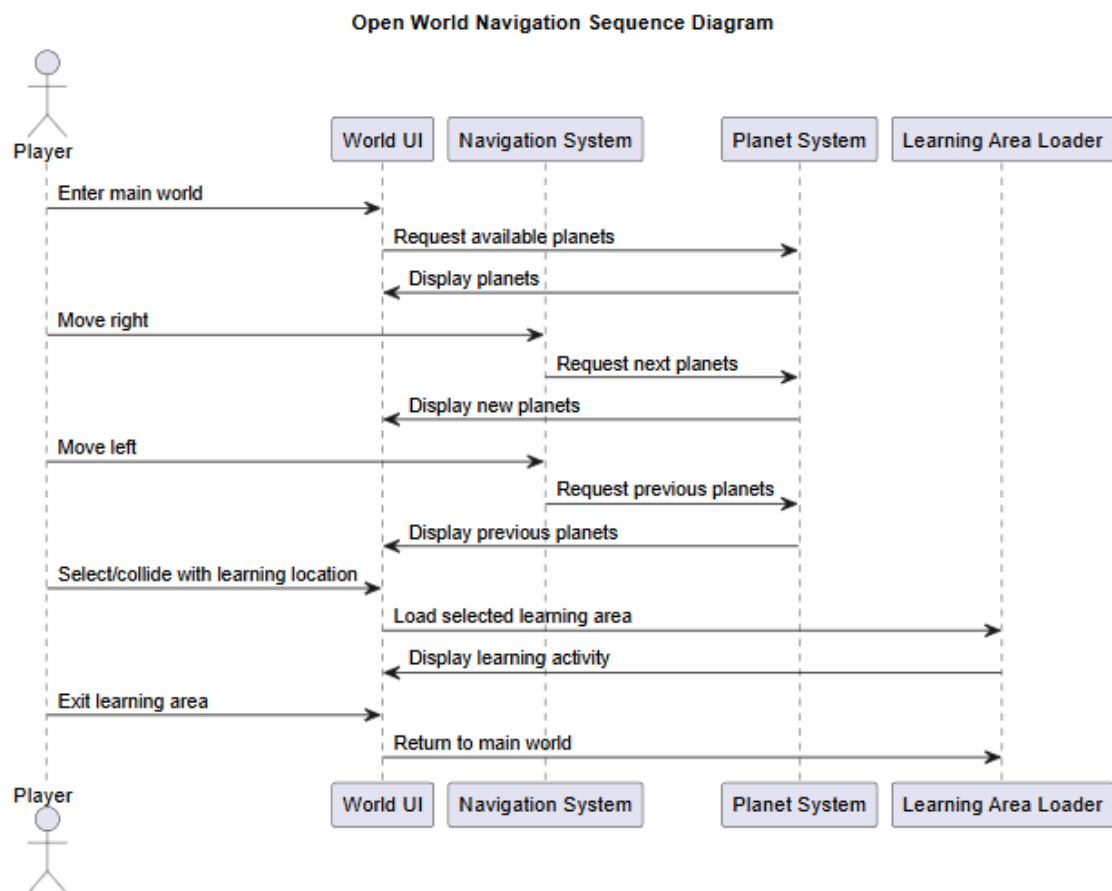


Figure 7: Open World Navigation Sequence Diagram

Table 9: Feedback Points and Motivation Sequence Diagram details

Item	Details
Actors	Player
Description	Player receives feedback after performing signs and earns points/stars for correct answers.
Data	Validation result, points, stars, progress record, feedback message
Stimulus	Player completes a sign or game task.
Response	The system gives positive or supportive feedback and updates the score/progress.
Comments	Motivates the player while keeping all content available without locking levels.

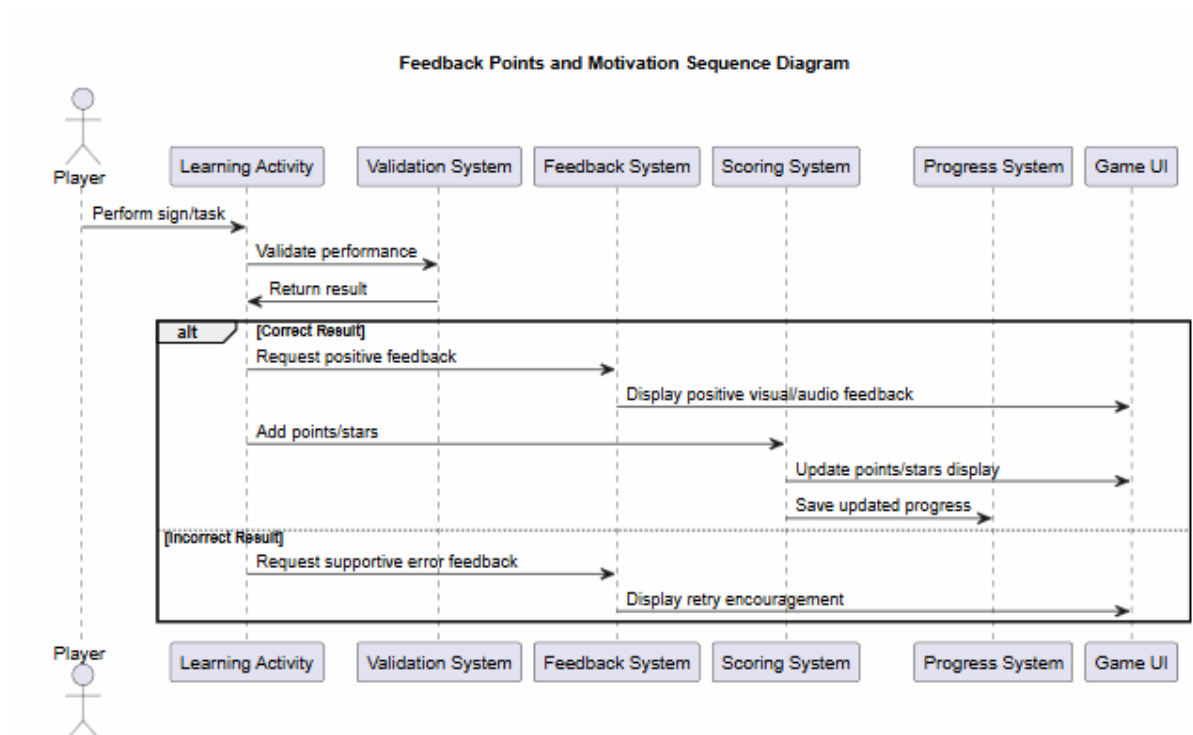


Figure 8: Feedback Points and Motivation Sequence Diagram

Table 10: Exit and Progress Saving Sequence Diagram details

Item	Details
Actors	Player
Description	Player exits the game, and the system saves and restores progress when the game is reopened.
Data	Player progress, learned signs, earned points/stars, selected avatar
Stimulus	Player exits or restarts the game.
Response	The system saves progress and restores it during the next session.
Comments	Ensures continuity of the learning experience.

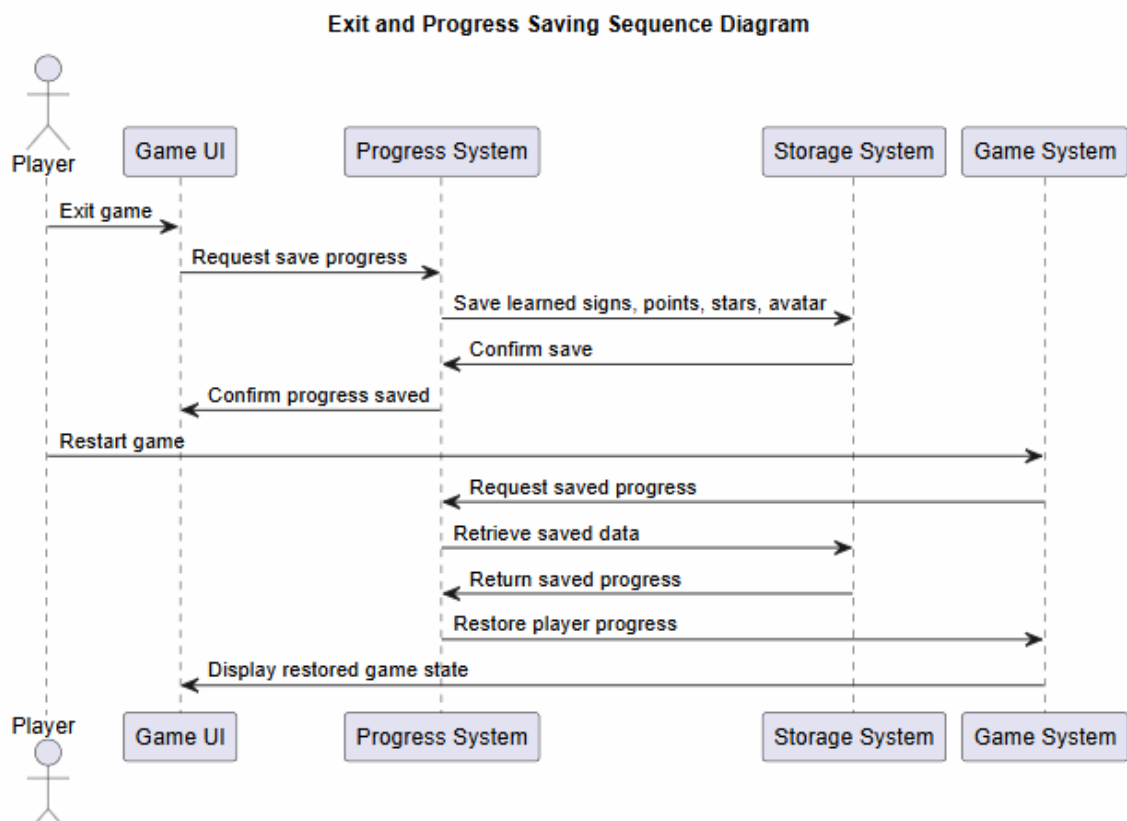


Figure 9: Exit and Progress Saving Sequence Diagram

Table 11: Sign Language Learning Interaction Sequence Diagram details

Item	Details
Actors	Player
Description	Player learns a target sign by viewing its demonstration, performing it using hand gestures, and receiving validation from the system.
Data	Target sign, sign demonstration, camera frames, hand landmarks, gesture result
Stimulus	Player enters a sign learning activity and performs the required sign.
Response	The system captures the gesture, compares it with the target sign, and determines whether it is correct or incorrect.
Comments	Represents the core AI-based learning interaction in SigniVerse.

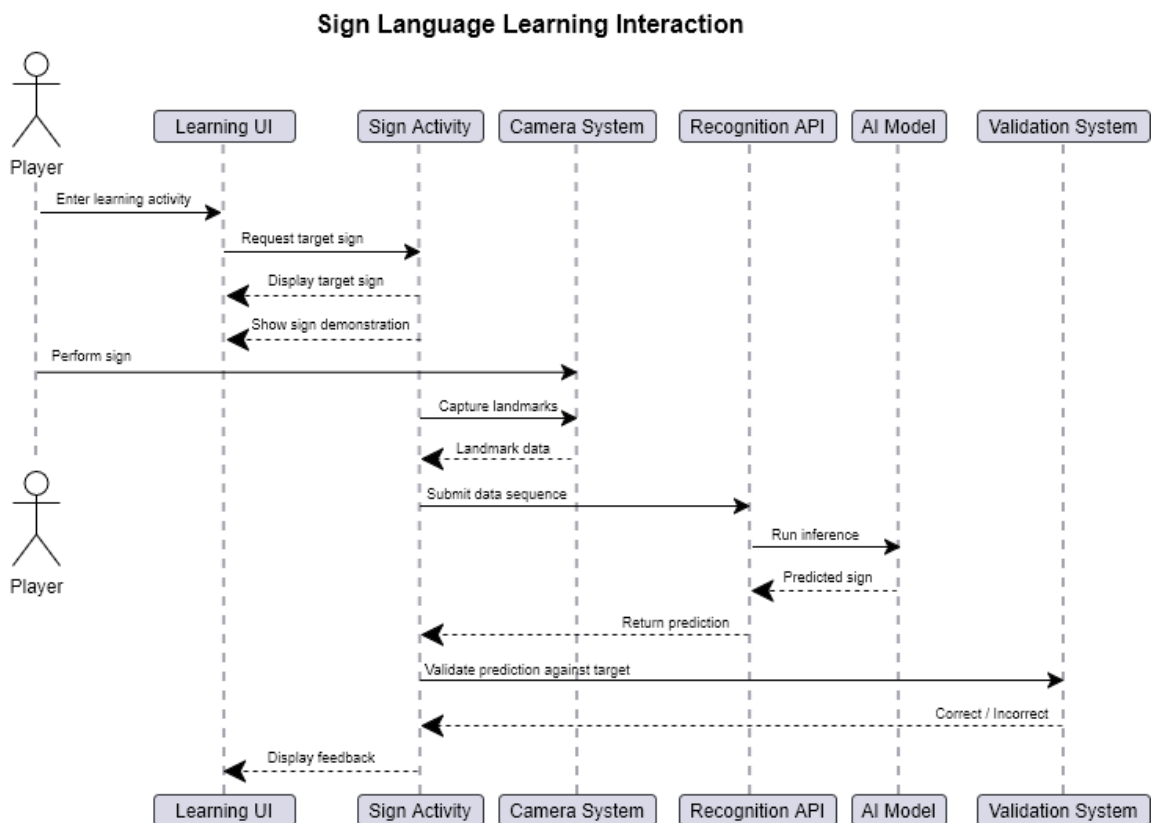


Figure 10: Sign Language Learning Interaction Sequence Diagram

Table 12: Help and Error Handling Sequence Diagram details

Item	Details
Actors	Player
Description	Player can retry the same sign multiple times after an incorrect attempt.
Data	Attempt number, target sign, incorrect result, retry status
Stimulus	Player performs an incorrect sign or chooses to try again.
Response	The system keeps the same sign active and allows another attempt.
Comments	Supports learning without punishment and encourages practice.

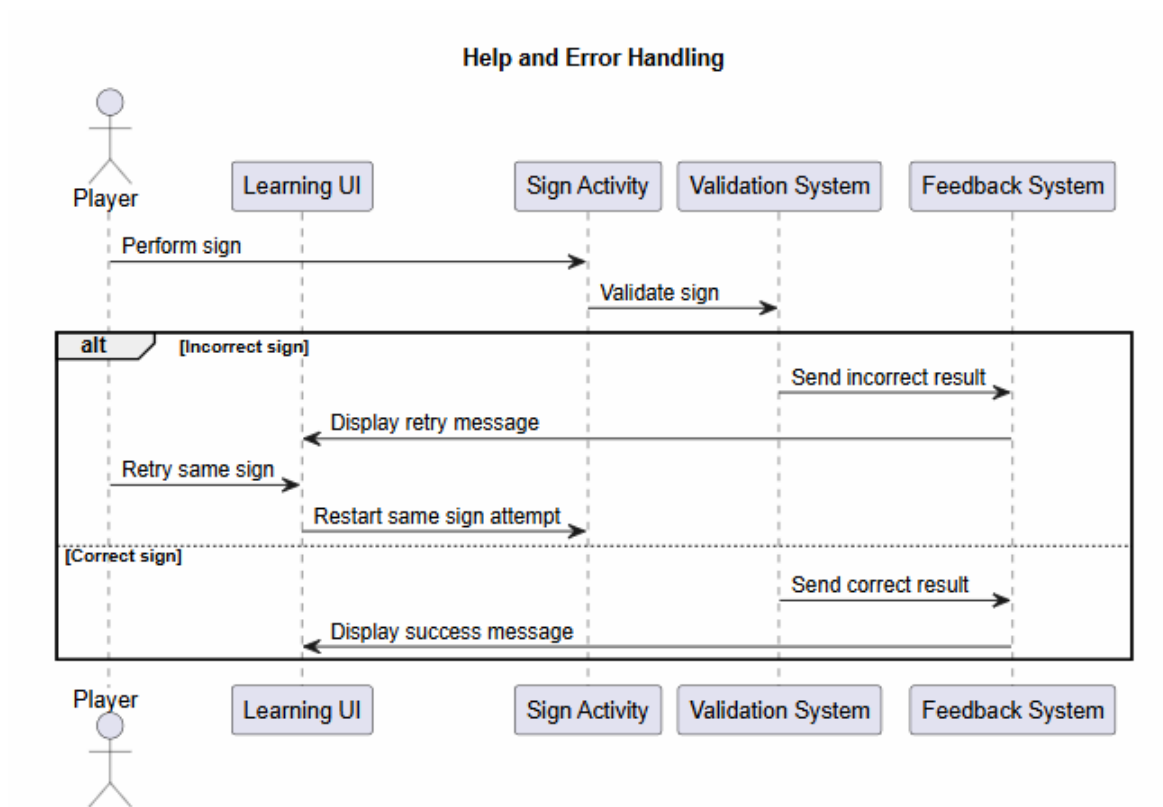


Figure 11: Help and Error Handling Sequence Diagram

Table 13: Shop and Avatar System Sequence Diagram details

Item	Details
Actors	Player
Description	Player opens the in-game shop, browses spacecraft avatars, purchases avatars using stars, and equips a selected avatar.
Data	Avatar list, avatar price, player stars, purchase status, selected avatar
Stimulus	Player clicks the shop button or selects an avatar.
Response	The system displays available avatars, checks star balance, completes purchase if possible, and updates the equipped avatar.
Comments	Adds motivation and personalization to the learning experience.

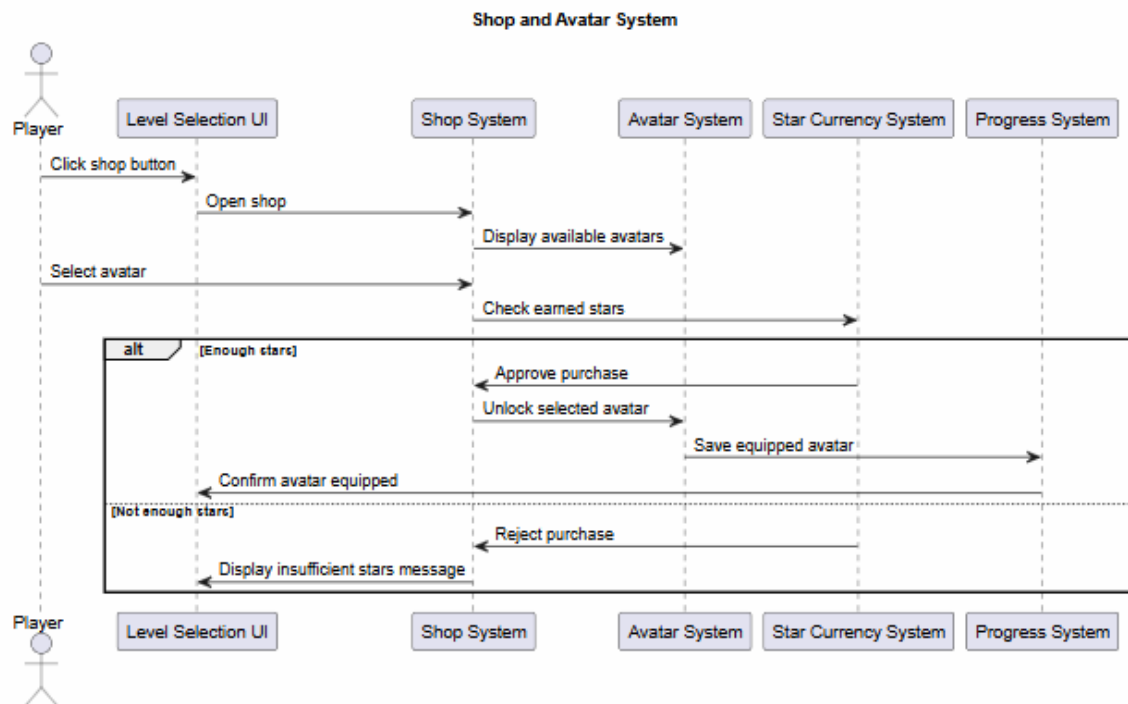


Figure 12: Shop and Avatar System Sequence Diagram

Table 14: Numbers Mini-Game Sequence Diagram details

Item	Details
Actors	Player
Description	Player participates in a numbers-related educational mini-game and receives a performance rating after completing the activity.
Data	Number targets, collected objects, score, stars, performance rating
Stimulus	Player starts the numbers mini-game and interacts with game objects.
Response	The system tracks object collection, evaluates performance, and awards stars.
Comments	Reinforces number learning through interactive gameplay.

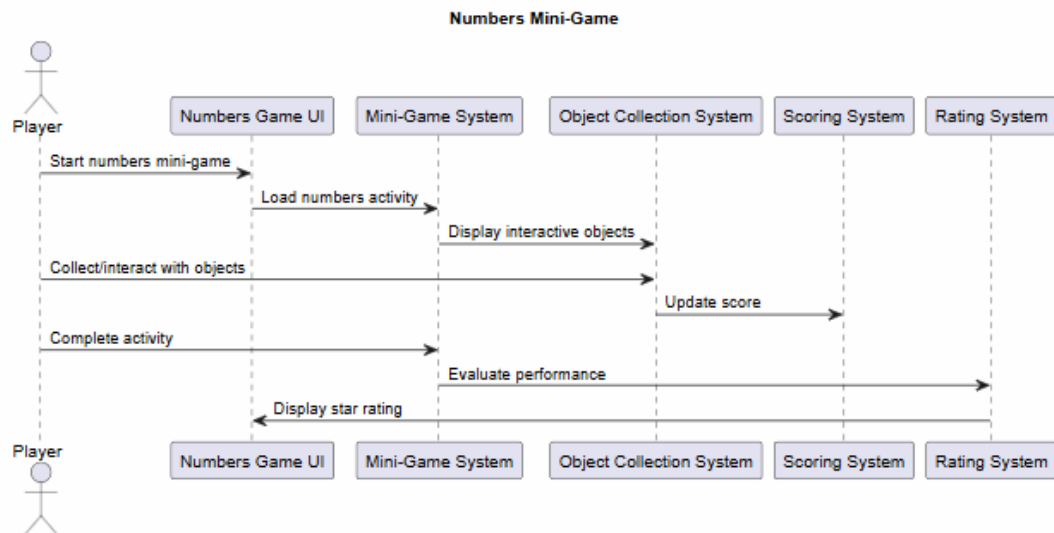


Figure 13: Numbers Mini-Game Sequence Diagram

Table 15: Alphabets Mini-Game Sequence Diagram details

Item	Details
Actors	Player
Description	Player controls a spaceship, avoids obstacles, collects stars, and progresses through an alphabet-related game activity.
Data	Spaceship position, obstacles, collectible stars, score, movement input
Stimulus	Player starts the alphabets mini-game and controls the spaceship.
Response	The system updates spaceship movement, detects obstacles, collects rewards, and displays the score.
Comments	Supports alphabet learning through action-based gameplay.

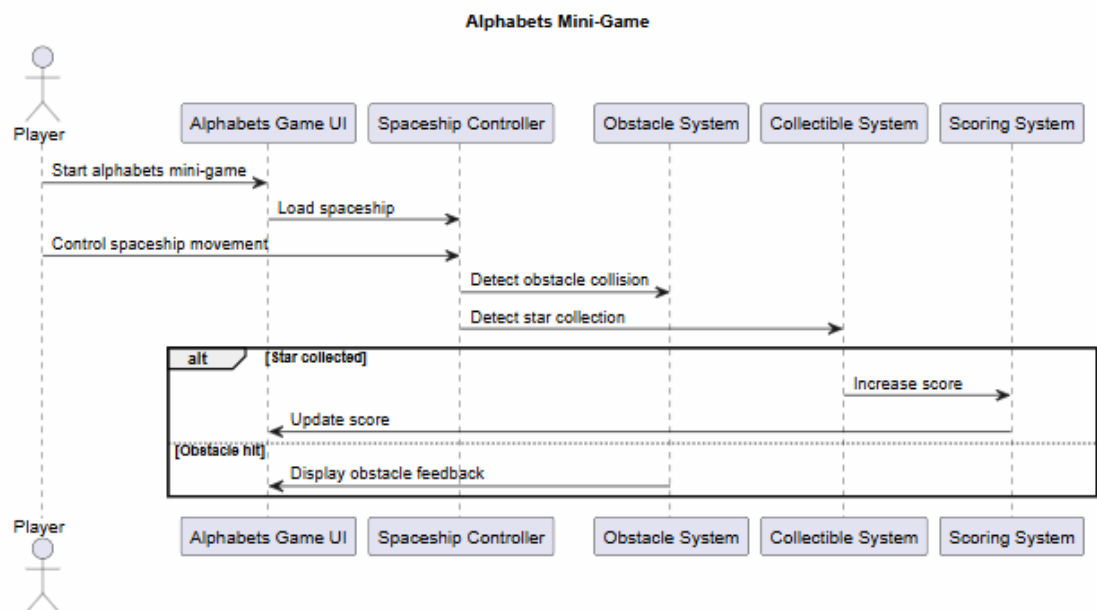


Figure 14: Alphabets Mini-Game Sequence Diagram

Table 16: Home World Puzzle Mini-Game Sequence Diagram details

Item	Details
Actors	Player
Description	Player completes a home puzzle activity based on four rooms: Bedroom, Living Room, Bathroom, and Kitchen.
Data	Puzzle pieces, room names, room positions, placement result, completion status
Stimulus	Player enters the Home World and places puzzle elements in the house.
Response	The system validates room placement, gives feedback, and confirms puzzle completion.
Comments	Reinforces home vocabulary using a puzzle-based learning activity.

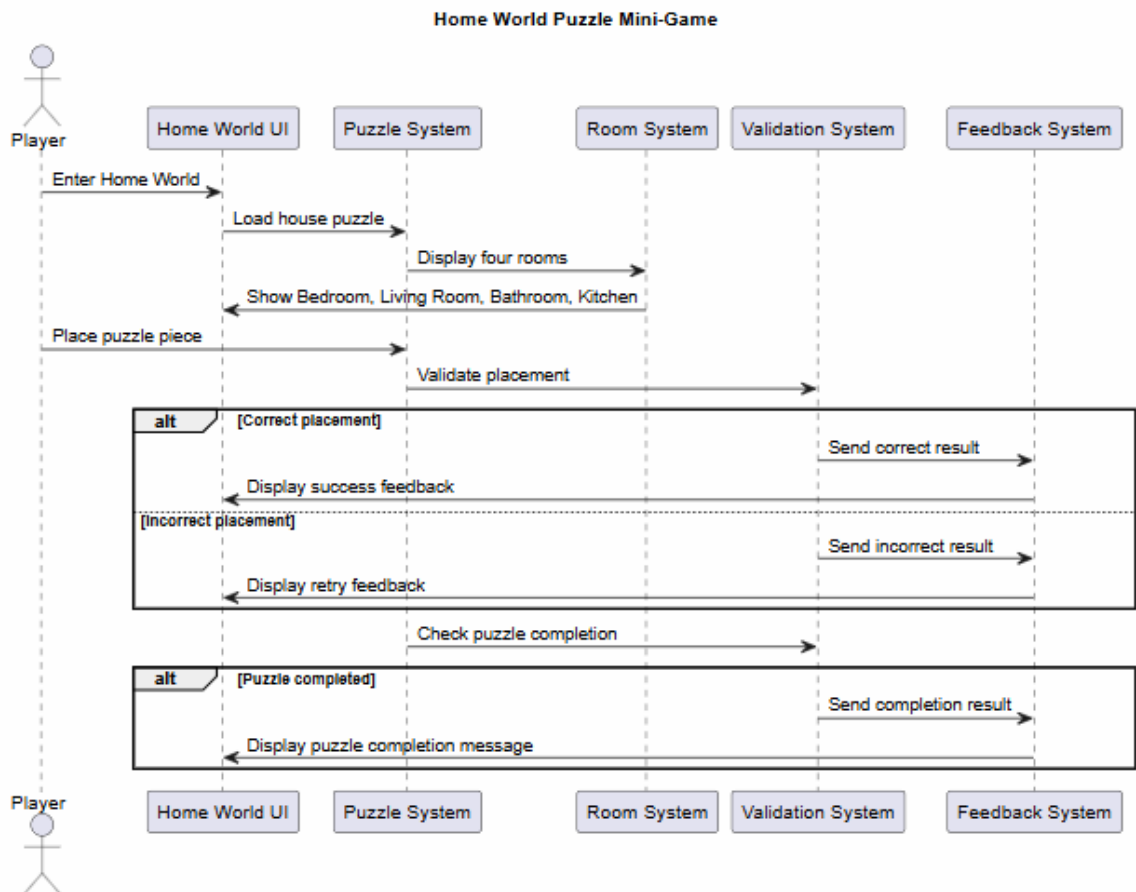


Figure 15: Home World Puzzle Mini-Game Sequence Diagram

4.5 UML Class Diagram

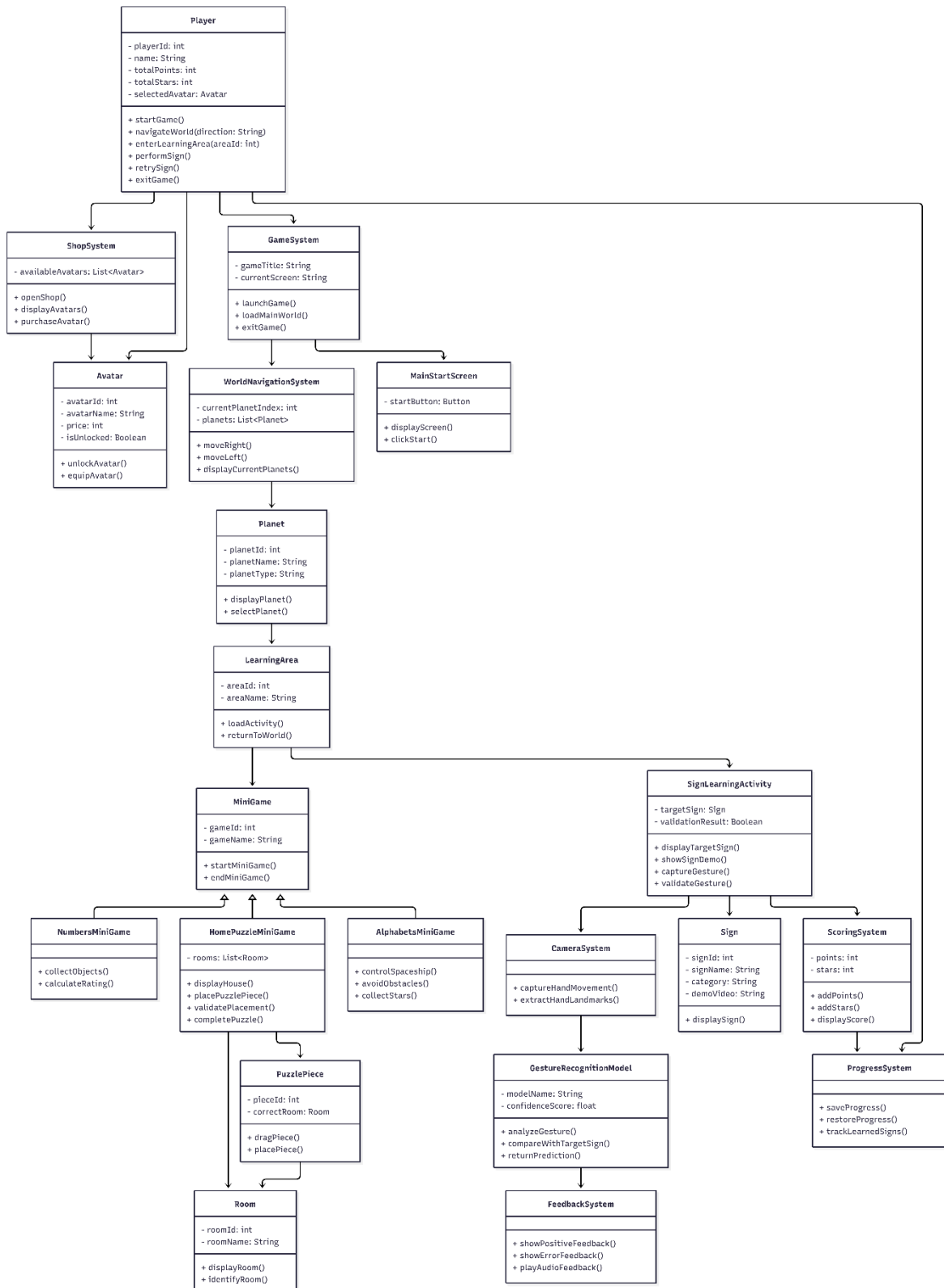


Figure 16: UML Class Diagram

4.6 Graphical User Interface

1. Main menu and home screen

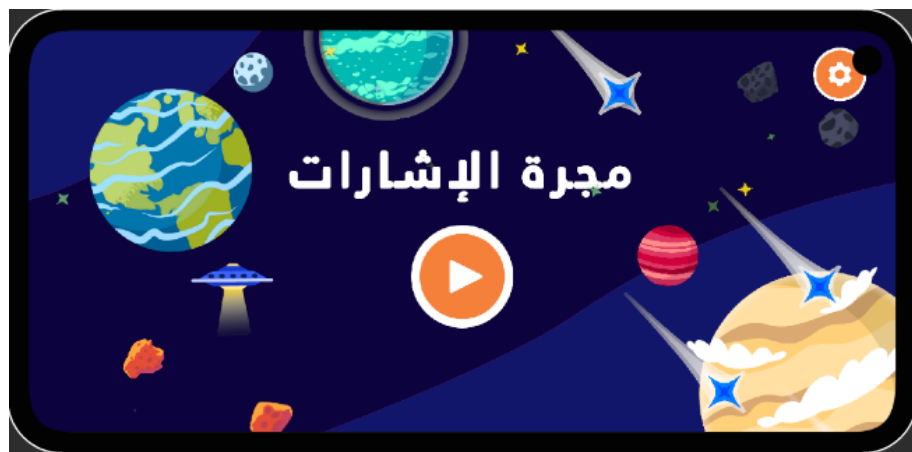


Figure 17: Home screen GUI



Figure 18: Main menu GUI

2. Avatar shop



Figure 19:avatar selection



Figure 20:avatars shop GUI

3. Sign performance & feedback

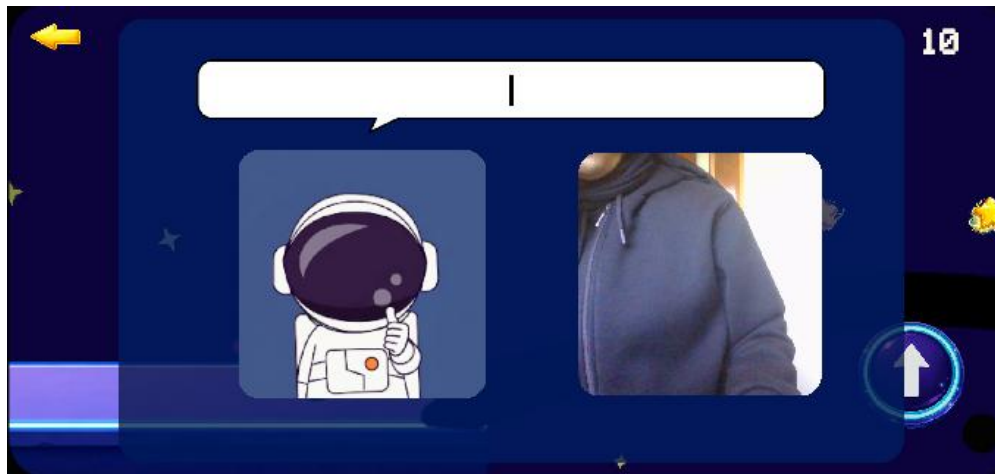


Figure 21: Sign to perform GUI

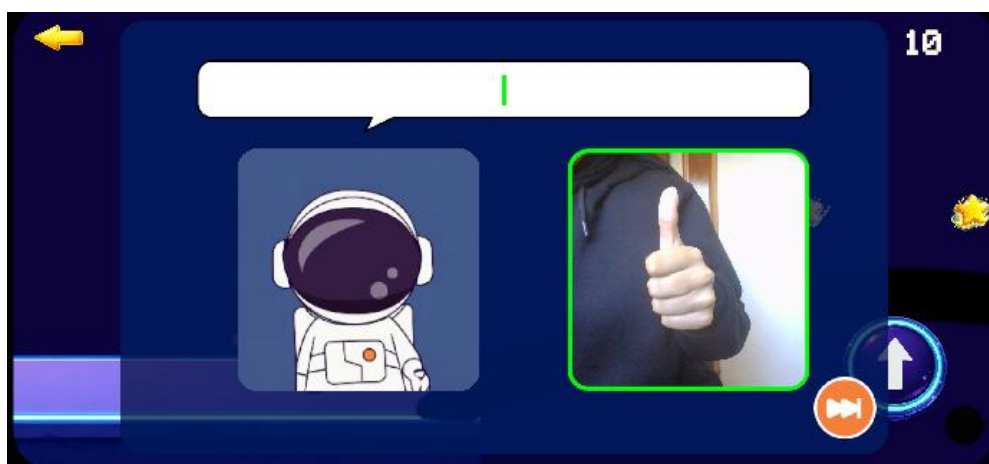


Figure 22: Sign feedback GUI

4. Numbers game



Figure 23: Collect stars GUI

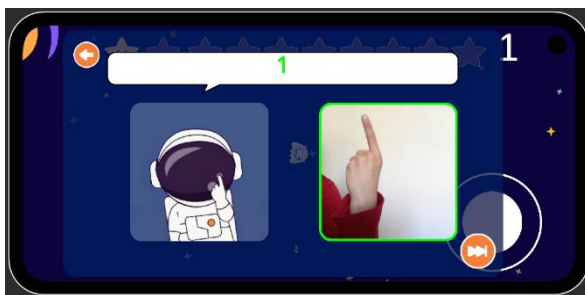


Figure 24: Sign number of collected stars



Figure 25: Winning panel after collecting 10 stars GUI



Figure 26: Cancellation panel GUI

5. Alphabets game

In the alphabet game, when the user reaches a bridge, the sign panel pops up.



Figure 27: Alphabet level selection



Figure 28: Alphabet gameplay GUI

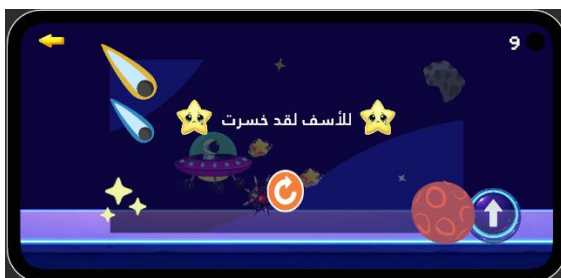


Figure 29: Losing GUI



Figure 30: Winning GUI

6. Home puzzle game

In the alphabet game, when the user reaches a bridge, the sign panel pops up.



Figure 31: Category selection GUI



Figure 32: Living room category GUI



Figure 33: Bedroom category GUI



Figure 34: Bathroom category GUI



Figure 35: Kitchen category GUI



Figure 36: Puzzle example GUI

5.0 SYSTEM IMPLEMENTATION

5.1 Implementation Tools and Environment

This section describes the software tools, and frameworks used to develop and deploy the **Signiverse** application.

- **Game Engine: Unity (Version 6000.3.6f1 LTS)** was selected as the primary development platform for its robust support for 3D/2D game environments and its seamless integration with external AI plugins.
- **Programming Language: C#** was used to implement game logic.
- **AI & Computer Vision Frameworks:**
 - **MediaPipe (by Google):** Utilized for real-time hand landmark detection, extracting 21 3D hand coordinates from the webcam feed.
 - **LSTM (Long Short-Term Memory):** A specialized Recurrent Neural Network (RNN) architecture used to recognize and classify **sequential/dynamic gestures** by analyzing the movement patterns of landmarks over time.
 - **TensorFlow / TensorFlow Lite:** Used to train LSTM models.
- **Backend API Framework: FastAPI** was selected as the high-performance asynchronous web framework to build the API endpoint. It serves as the bridge between Unity and the machine learning model, receiving hand landmark coordinates sent over HTTP POST requests from the game client and returning real-time gesture classifications.
- **Hosting & Deployment Platform: Render** was utilized as the unified cloud platform to host and deploy the live FastAPI server. Render handles continuous deployment directly from the GitHub repository, providing a stable, production-ready URL that the Unity client targets during live gameplay.

- **Asset Generation & Animation Tools:** **Kling AI** was utilized to produce the instructional sign language content. The platform was used to animate a custom-designed astronaut avatar, mapping recorded sign language gestures onto the digital character to generate high-quality, fluid instructional video clips for the player.
- **Integrated Development Environment (IDE):** **Visual Studio** was used for writing and debugging the C# scripts.
- **Version Control:** **Git** and **GitHub Desktop** were utilized for collaborative development and version management.

5.2 Data Persistence (PlayerPrefs)

To manage user progress and achievements efficiently, the system utilizes Unity's **PlayerPrefs** system. This allows for lightweight and fast local data storage:

- **Progress Tracking:** Stores unlocked levels, high scores, and planet completion status.
- **Data Logic:** Uses a key-value pair mechanism to ensure that the child's progress is saved and loaded automatically every time the game is launched.

6.0 Game Testing and Installation

6.1 Introduction

This chapter presents the testing and installation process of the game. The purpose of testing is to ensure that the game works correctly, is stable, easy to use, and provides a good user experience. The chapter includes functional testing, usability testing, cooperative evaluation, installation steps, and a user manual that explains how to play the game.

6.2 Testing Environment

The game was tested on Unity Editor and on the target device to make sure that all gameplay features work correctly. The testing focused on player movement, jumping, collecting stars, losing, winning, level unlocking, and navigation between game screens.

Table 17: Testing Environment

Item	Description
Game Engine	Unity
Programming Language	C#
Testing Device	Laptop / Android Phone
Operating System	Windows / Android
Input Method	Keyboard / Touch / Motion detection
Build Type	APK / Windows Build / Unity Editor
Testers	Students / Friends / Family members

6.3 Functional Testing

Table 18:Functional Testing

Test Case ID	Feature	Test Steps	Expected Result	Actual Result	Status
TC1	Start Game	Open the game and press the button Start	The planets page opens	Planets page opened	Pass
TC2	Numbers mini game	Press on the numbers planet in planets page	The numbers game opens	Numbers game opened	Pass
TC3	Joystick	Move it in different directions	The character moves in the same directions	The character moved in the same directions	Pass
TC4	Collect Star	Touch/collide with a star	Star count increases and the panel opens to do the sign	Star count increased and the panel opened to do the sign	Pass
TC5	Win panel	Collect all 10 stars	Win panel opens	Win panel opened	Pass
TC6	Alphabets mini game	Press on the alphabets planet in planets page	The alphabets game opens	The alphabets game opened	Pass
TC7	Player Movement	Use the control keys/buttons	The player moves correctly	Player moved correctly	Pass
TC8	Jump	Press jump button	The player jumps once when on ground	Jump worked	Pass
TC9	Collect Stars	Touch/collide with a star	Star count increases	Star count increased	Pass

TC10	Obstacle Collision	Hit an obstacle	Lose panel appears	Lose panel appeared	Pass
TC11	Restart Level	Press Restart after losing	Same level restarts	Level restarted	Pass
TC12	Win Level	Reach finish line	Win panel appears	Win panel appeared	Pass
TC13	Next Level	Press next level	Next level opens	Next level opened	Pass
TC14	Level Unlocking	Win Level 1	Level 2 becomes unlocked	Level 2 unlocked	Pass
TC15	Back button	Press back button	Player returns to main menu	Main menu opened	Pass
TC16	Home mini game	Press on the home planet in planets page	The home game opens	The home game opened	Pass
TC17	Open bedroom	Press on bedroom	The room opens	The room opened	Pass
TC18	Open bathroom	Press on bathroom	The room opens	The room opened	Pass
TC19	Open living room	Press on living room	The room opens	The room opened	Pass
TC20	Open kitchen	Press on kitchen	The room opens	The room opened	Pass
TC21	Rooms items	Press any item	A puzzle for this item will open	The puzzle opened	Pass
TC22	Market button	Press market button on planets page	The market opens	The market opened	Pass
TC23	Buy character	Press buy on market	The character purchase	The character purchased	Pass

6.4 Heuristic Evaluation

Heuristic evaluation was conducted to check the usability of the game interface. The evaluation focused on whether the game screens, buttons, feedback messages, and navigation are clear and easy to use.

Table 19:Heuristic Evaluation

No.	Heuristic	Description for the Game
H1	Visibility of System Status	The game should show score, stars, level status, win/lose feedback clearly.
H2	Match Between Game and Real World	Icons, buttons, and instructions should be familiar and easy to understand.
H3	User Control and Freedom	The player should be able to restart, or return to menu easily.
H4	Consistency and Standards	Buttons, colors, fonts, and panels should be consistent across all levels.
H5	Error Recovery	The game should show clear feedback when the player loses or cannot continue.
H6	Error Prevention	The game should prevent unfair losing, repeated jumps, or broken level transitions.
H7	Recognition Rather Than Recall	Controls and objectives should be visible instead of making the player guess.
H8	Flexibility and Efficiency	Players should be able to restart or move to the next level quickly.
H9	Aesthetic and Minimalist Design	The game UI should be simple and not crowded.
H10	Help and Documentation	The game should provide instructions or a short tutorial.

Summary table

Table 20: Summary table

Heuristic	Issues Found	Severity
H1	1	Minor
H4	1	Minor
H5	1	Minor
H7	2	Minor
H10	1	Minor

6.5 Cooperative Evaluation

Cooperative evaluation was performed with three participants. Each participant was asked to play the game and complete a set of tasks while speaking about any difficulties they faced. The goal was to identify usability problems and measure how easy the game is to understand.

Table 21: Cooperative Evaluation

Criteria	Participant 1	Participant 2	Participant 3	Participant 4
Gender	Female	Male	Female	Male
Age	9 years old	8 years old	18 years old	22 years old
Educational Level	3 rd grade	2 nd grade	University Student	University Student
Gaming Experience	Medium	Beginner	Medium	Medium

Tasks Table

Table 22:Tasks Table

Task No.	Task	Participant 1 Time	Participant 2 Time	Participant 3 Time	Participant 4 Time	Comments
1	Start the game	4 sec	6 sec	3 sec	3 sec	-
2	Understand the controls	17 sec	20 sec	13 sec	13 sec	-
3	Collect a star	12 sec	15 sec	12 sec	11 sec	-
4	Avoid obstacle	19 sec	25 sec	17 sec	17 sec	-
5	Restart after losing	4 sec	8 sec	3 sec	4 sec	-
6	Win the level	60 sec	66 sec	49 sec	50 sec	-
7	Go to next level	10 sec	11 sec	8 sec	9 sec	-
8	Return to main menu	5 sec	6 sec	4 sec	5 sec	-
9	Open the market	5 sec	7 sec	5 sec	5 sec	-
10	Buy character	10 sec	13 sec	9 sec	10 sec	-

6.6 Post-Test Questionnaire

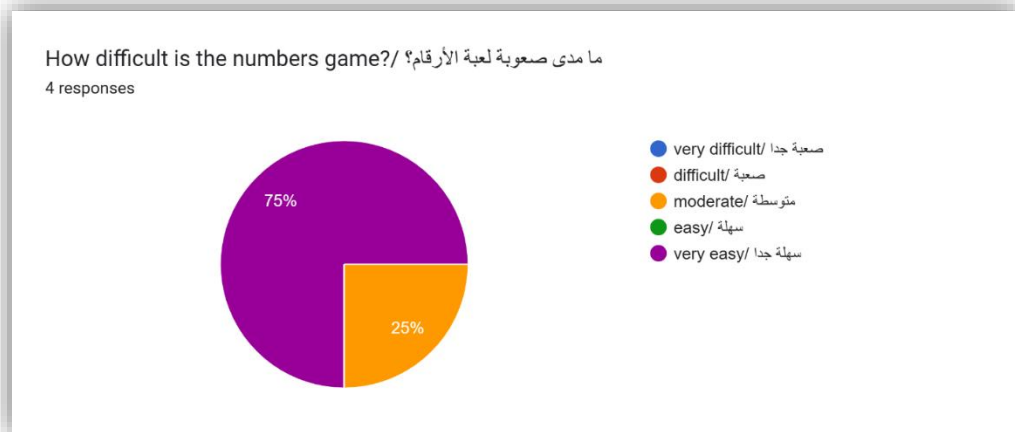


Figure 37: Post-Test Questionnaire

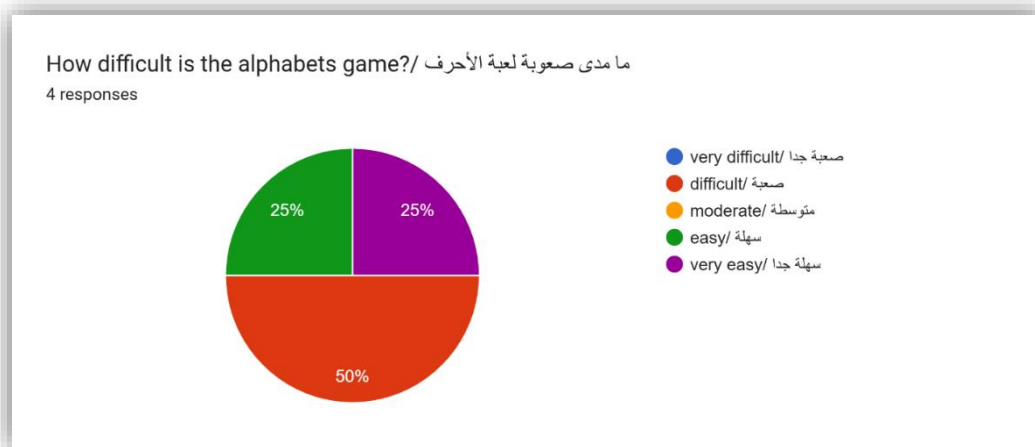


Figure 38: Post-Test Questionnaire

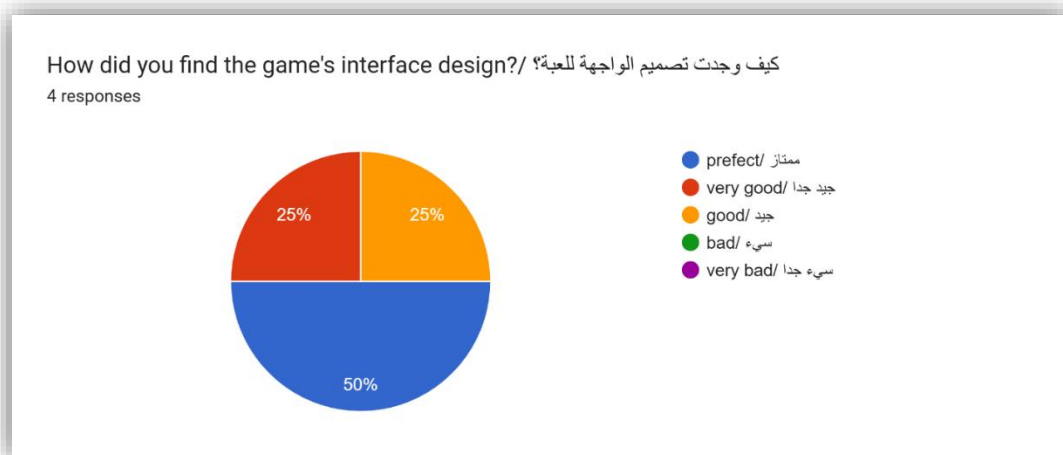


Figure 39:Post-Test Questionnaire



Figure 40:Post-Test Questionnaire

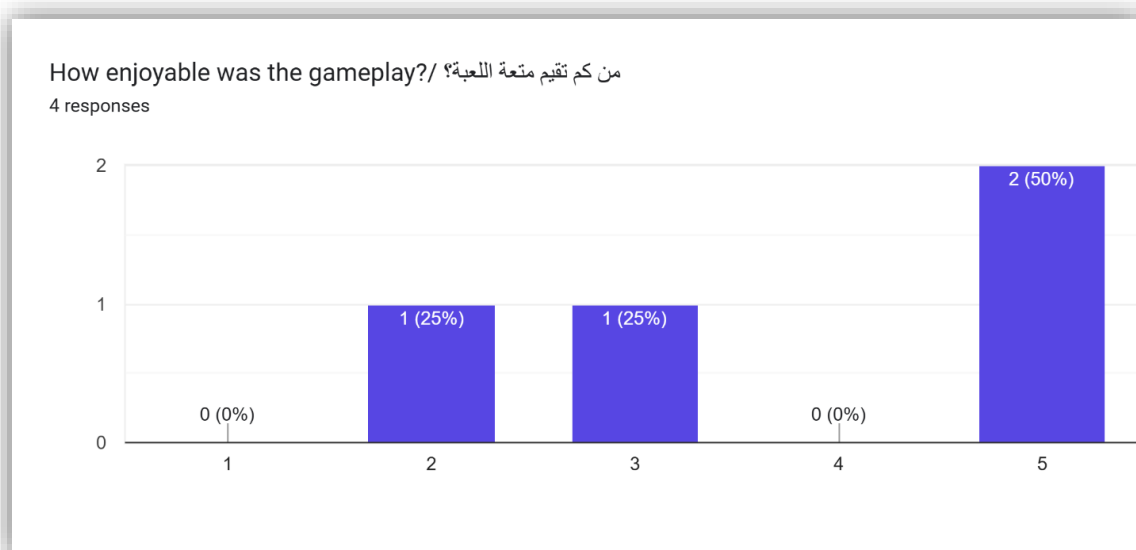


Figure 41:Post-Test Questionnaire

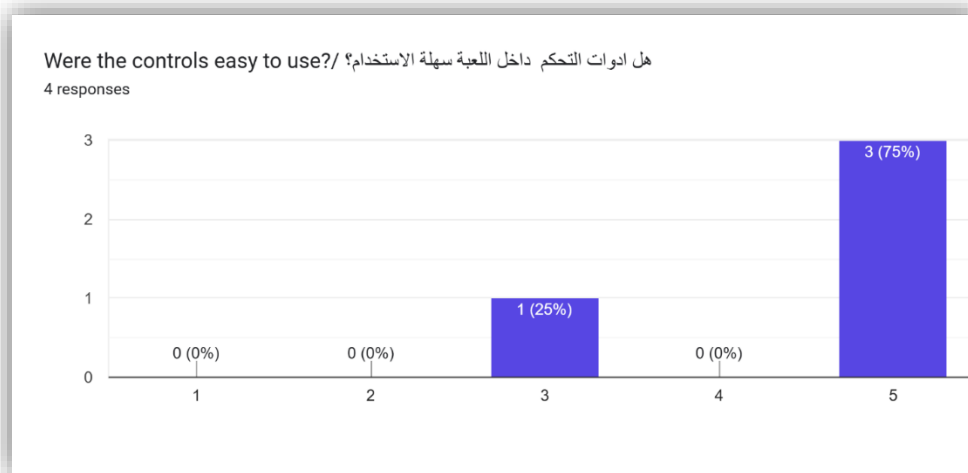


Figure 42: Post-Test Questionnaire

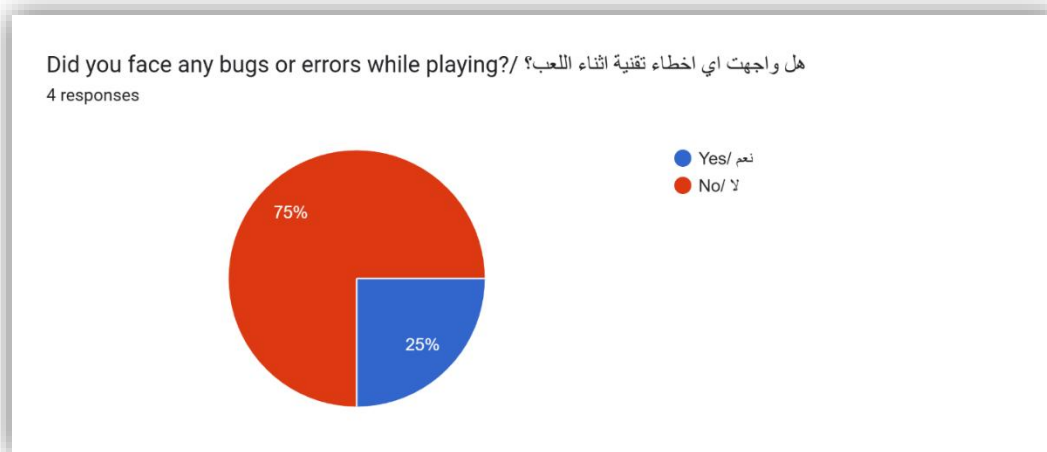


Figure 43: Post-Test Questionnaire

If yes, please describe the bug or problem / اذا كانت الاجابة نعم اشرح الخطأ الذي واجهك

1 response

نظرت عن خط النهاية وبالتالي ما فاز بلعبة الاحرف

Figure 44: Post-Test Questionnaire

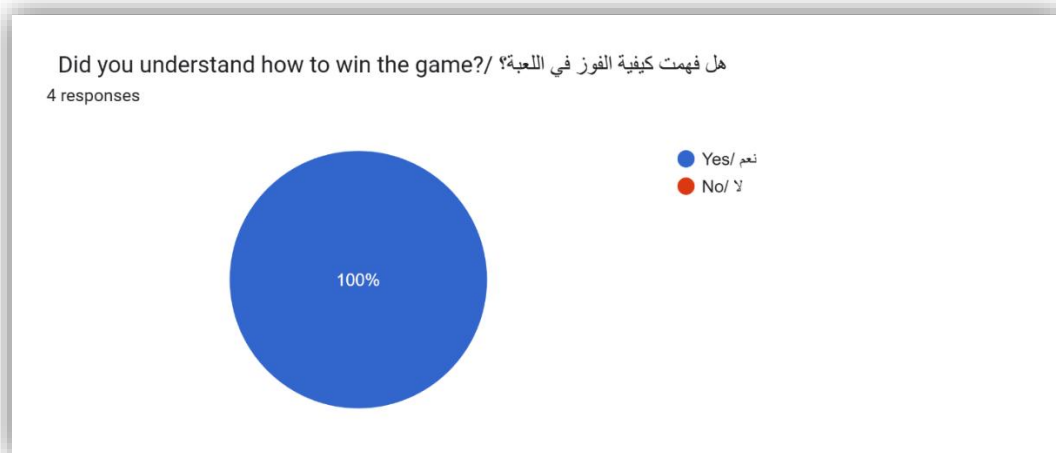


Figure 45: Post-Test Questionnaire

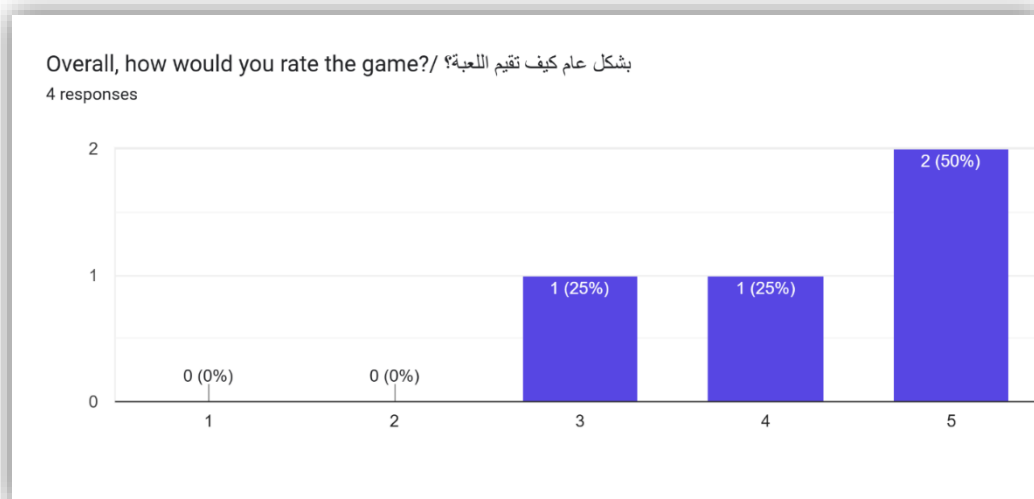


Figure 46: Post-Test Questionnaire

Any additional comments or suggestions? / اي تعليق اضافي او اقتراحات؟

4 responses

انه لازم ننشرها عشان تعلم الاطفال

No

"كنتا عمرها 6 بتحكليكم اللعبة "مزهقة

الشخصية الصغيرة التي بتتحرك صعب الواحد يتخطاها (لعبة الأحرف)

Figure 47: Post-Test Questionnaire

6.7 System Installation

This section explains how to install the game on an Android device.

Table 23: System Installation

Requirement	Description
Device	Android phone
OS Version	Android 8.0 or higher
Storage	At least 200 MB free space
Internet	required

Installation Steps:

1. Download the game APK file.
2. Open the APK file on the Android device.
3. If required, allow installation from unknown sources.
4. Press Install.
5. Open the game from the app list.
6. Start playing from the main menu.

6.8 User Manual



Figure 48: Main Menu Screen

The main menu is the first screen displayed when the game starts. It contains the game title “مجرة الإشارات” and a play button in the center of the screen. The user can press the play button to start the game. The settings icon is also available in the top-right corner to access game settings.



Figure 49 Learning Categories Screen

After pressing the play button, the user is directed to the learning categories screen. This screen displays the available learning paths such as **numbers**, **letters** and **home**. The user can choose a category to start learning and playing levels related to that topic.



Figure 50: Character Selection Screen

This screen allows the user to choose a playable character. The available characters are shown at the bottom of the screen. After selecting a character, the user can press the green “المتجر” button to open the store or continue with the selected character.



Figure 51: Store Screen

The store screen displays different characters that the player can buy using collected stars. Each character has a price. If the player has enough stars, they can press the “شراء” button to buy the character. If the player does not have enough stars, a message such as “No Enough Coins” appears.



Figure 52: Numbers Learning Screen

This is the numbers game, where the joystick can be used to collect stars and learn numbers. The player can see collected stars at the top of the screen, with the number displayed.

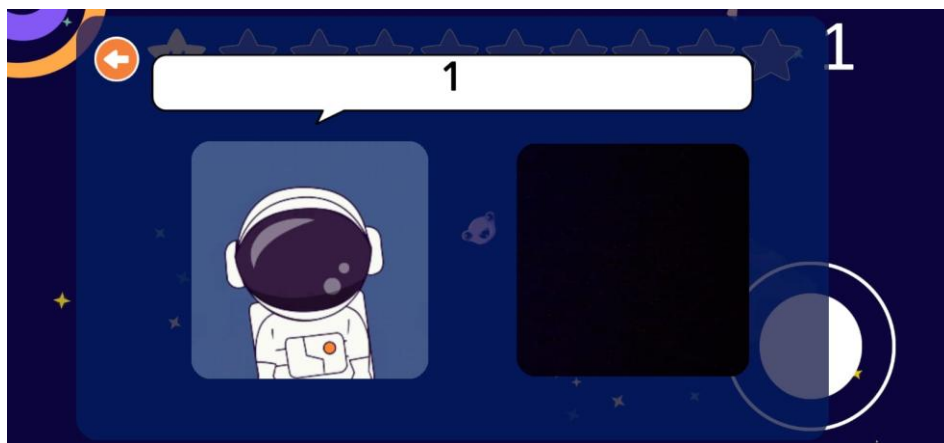


Figure 53: sign panel

During the game, a panel appears on the screen. The panel displays video for the sign, and the player must do the same sign as the video. This helps the user practice recognizing signs and connecting them with the correct meaning.



Figure 54: Alphabets levels screen

The alphabets level screen displays all available levels using star-shaped buttons. Unlocked levels appear clearly, while locked levels appear darker. The player can choose any unlocked level to start playing. Locked levels become available after completing previous levels successfully.



Figure 55: Alphabets game screen

The player controls the character and moves through the space environment. The player can press the jump button on the right side of the screen to avoid obstacles and collect stars. The score is displayed at the top-right corner, and the back button is available at the top-left corner.



Figure 56: Lose panel

This screen appears when the player loses the level, for example when the player hits an obstacle. The message “للأسف لقد خسرت” is displayed to inform the player that the level was not completed successfully.

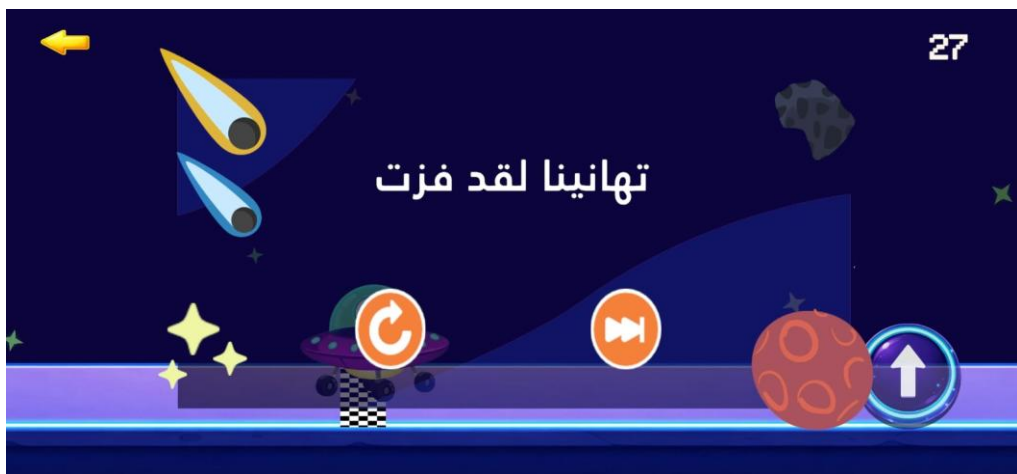


Figure 57: Win panel

This screen appears when the player successfully completes the level. The message “تهانينا لقد فزت” is displayed to inform the player that the level has been completed successfully. The player’s current score is shown at the top-right corner of the screen. The screen also provides two main options: the **restart** button, which allows the player to replay the current level, and the **next** button, which allows the player to move to the next level.

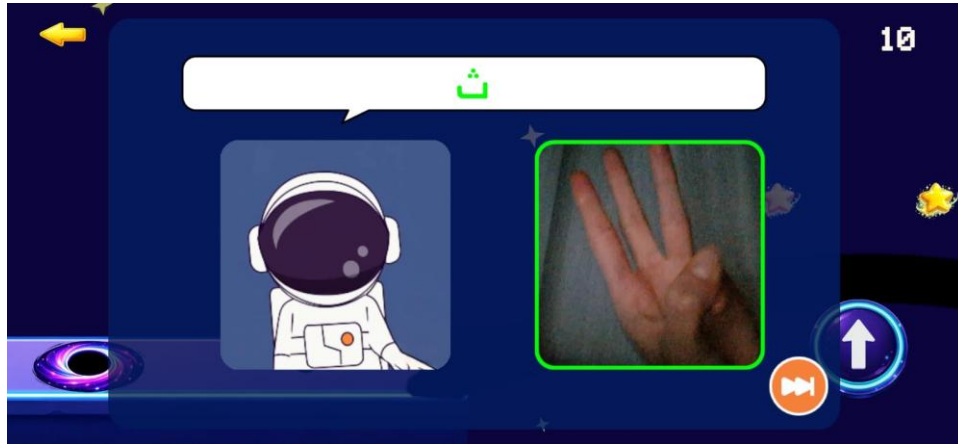


Figure 58: The sign detected correctly

The camera area shows the player's hand gesture, and the model checked that the sign is correct.



Figure 59: Exit Confirmation panel

When the player tries to leave the level before completing it, a confirmation message appears. The message asks the player if they are sure about leaving because they may lose the stars related to the current sign. The user can press “إلغاء” to stay in the level or “نعم، أخرج” to exit.



Figure 60: Home game

When the player enters the Home Objects mini-game, a full house screen appears. The house is divided into different rooms such as the bedroom, bathroom, living room, and kitchen. The player can select any available room by tapping on it. Each room contains a group of objects related to that room. The back arrow at the top-left corner allows the player to return to the previous screen.



Figure 61: Bedroom

The bedroom screen includes objects such as a bed, wardrobe, window, and folded clothes. The player taps on an object to start the related learning activity. This screen supports vocabulary learning for items commonly found in a bedroom.



Figure 62: Living room

After selecting the living room, the player sees several objects that belong to the living room, such as a chair, table, television, phone, lamp, and cabinet. The player taps on one object to start playing puzzle game. This screen helps the child connect the object with its real-life location inside the house.



Figure 63: Kitchen

The kitchen screen displays common kitchen tools and objects, such as the refrigerator, spoon, plate, cup, and other kitchen items. The player chooses one object by tapping on it. After selecting the object, the game moves to the puzzle activity related to that object.



Figure 64: Puzzle game

After choosing an object, the player is directed to a puzzle screen. The selected object appears as a faded image in the center, and puzzle pieces appear around it. The player must drag or place the puzzle pieces in the correct positions to complete the object image. This activity helps the child focus on the object shape before learning its sign.



Figure 65: Sign panel

After completing the puzzle, the sign recognition screen appears. The name of the selected object is displayed at the top of the screen, such as “مغسلة”. The left side shows an astronaut performing the sign, while the right side displays the camera view. The player must perform the correct gesture in front of the camera to progress. If the sign is recognized correctly, the camera frame becomes highlighted, and the player can continue to the next step by pressing the next button.

7.0 PROJECT CONCLUSIONS AND FUTURE WORK

7.1 Project Conclusions

The primary objective of **Signiverse** was to bridge the educational gap for deaf children by creating an immersive, gamified environment that teaches sign language using artificial intelligence. By successfully integrating **Unity** with a temporal deep learning architecture (**MediaPipe and LSTM**), the project proves that real-time computer vision can be effectively utilized for interactive learning without the need for specialized hardware.

The system successfully tracks hand movements via a standard webcam, processes the sequential data to recognize dynamic signs, and updates the user's progress. Ultimately, Signiverse provides a safe, engaging, and motivating space-themed journey that transforms sign language from a rigid study into an interactive game.

7.2 Project Strengths and Weaknesses

7.2.1 Strengths

- **Gamified Learning Experience**

Signiverse utilizes a space-themed environment combined with a progressive reward system to create an engaging and enjoyable learning experience. This gamified approach helps maintain children's attention, increases motivation, and encourages continuous learning through interactive challenges and achievements.

- **Inclusivity and Universal Design**

Although the application is primarily designed for deaf children, its intuitive interface and easy-to-understand gameplay make it accessible to non-deaf children as well. This inclusive design promotes social integration, raises awareness of Jordanian Sign Language, and fosters empathy and communication between different groups of children.

- **Accessibility and Cost Efficiency**

The system operates using standard webcams and locally processed computer vision technologies, eliminating the need for specialized motion-tracking equipment. This significantly reduces deployment costs and makes the platform accessible to a broader range of users.

- **Socioeconomic Accessibility**

Signiverse addresses a significant educational challenge by providing a free and home-based learning solution. It helps overcome barriers associated with the high costs of private tutoring and the limited availability of specialized educational centers for deaf children, ensuring wider access to sign language education.

7.2.2 Weaknesses (Limitations)

- **Environmental Dependencies**

The accuracy of the AI model is highly dependent on ambient lighting conditions and camera quality. Low-light environments can reduce MediaPipe's landmark tracking precision.

- **Local Storage Restriction**

Since data is stored via PlayerPrefs on the local device, user progress cannot be synced or accessed from a different machine or platform.

7.3 Future Work

To elevate **Signiverse** into a commercial-grade educational tool, the following future enhancements are proposed:

- **Expanding the Vocabulary Dataset**

Training the LSTM model on a wider range of signs, including full sentences.

- **Cross-Platform Deployment and Mobile Distribution**

Future development will focus on deploying Signiverse across multiple platforms, including Android and iOS devices. Publishing the application on the **Google Play Store**

and **Apple App Store** would significantly increase accessibility and allow learners to access the platform from smartphones and tablets. This expansion would help reach a larger audience and provide a more flexible learning experience regardless of the device being used.

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